

Week 7

Name:

Mobile:

Personal Development Workouts

1. Watch the movie “Coach Carter”

Write a short description about this task

Link to the folder containing your audio summary

Technical Workouts

1. Database Administration
 - a. Monitor the database:
 - i. Check database status and current operations.
 - ii. Analyze query performance using explain plans.
 - iii. Enable profiling to identify slow queries.
 - b. Perform essential maintenance tasks:
 - i. Rotate logs for storage management.
 - ii. Upgrade the database to the latest version.
 - iii. Copy or clone databases for backup or testing purposes.
 - c. Manage users and roles:
 - i. Create and manage users, roles, and permissions.
 - d. Optimize configuration:
 - i. Modify MongoDB configuration files for improved performance.
2. Backup and Security
 - a. Implement backup strategies:
 - b. Use mongoexport and mongoimport for data export/import.
 - c. Perform binary backups with mongodump and mongorestore.

- d. Explore advanced backup methods:
 - i. Oplog backups.
 - ii. Logical Volume Manager (LVM) snapshots.
 - iii. Use MongoDB Management Service (MMS) or Ops Manager.
- e. Strengthen security:
 - i. Set up authentication and authorization for access control.
 - ii. Apply Role-Based Access Control (RBAC).
 - iii. Configure encryption and secure MongoDB instances.

3. Replication

- a. Set up replication:
 - i. Understand and configure Replica Sets.
 - ii. Learn the role of the Oplog in replication.
- b. Explore advanced replication configurations:
 - i. Set up Arbiter nodes for election voting.
 - ii. Use Hidden and Delayed nodes for analytics and recovery.
 - iii. Configure node priorities and manage elections.
 - iv. Resync nodes and handle rollbacks.
- c. Secure replication:
 - i. Enable keyfile authentication to secure node communication.

4. Scalability

- a. Understand scaling approaches:
 - i. Implement vertical scaling by increasing node resources.
 - ii. Explore horizontal scaling to distribute data across multiple nodes.
- b. Practice sharding:
 - i. Learn components of sharding: Shard Key, Chunks, Config Servers, and Query Routers.
 - ii. Compare types of sharding and choose an appropriate shard key.
- c. Follow sharding best practices:
 - i. Ensure balanced data distribution across shards.
 - ii. Decide between sharded and non-sharded collections.

iii. Implement tag-aware sharding and manage shard replicas.

Write a short description about this task

Miscellaneous Workouts

1. Practice typing for at least one hour each day on typing club, <http://typingclub.com/> . Finish as many chapters as possible as you can. Don't spend more than an hour each day.
2. Prepare a topic for the tech seminar. Record and upload it on youtube as an unlisted video. Topic eg: TCP/IP
3. Prepare and conduct a Feedback session by the end of this week.
4. Prepare your progress video for the last week. Record and upload it on youtube as an unlisted video.

Write a short description about this task

Link to screenshot image

Write a short description about this task

Link to your seminar video

Link to the document containing notes for your feedback session

Write a short description about this task

Link to your progress video